Can AI Describe Every Reptile?

A trait-extraction and knowledge-gap platform

VCU Computer Science Senior Design, 2026-2027

Prepared for: Dr. Bridget Thomson-McInnes (CS) and Dr. Peter Uetz (Biology; maintainer of the Reptile Database)

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Status: Working copy, restructured to follow the Project Preliminary Design template. Items in [brackets] are open decisions or placeholders.

Executive Summary

More than 12,000 reptile species are described by physical traits such as scale counts, coloration, and body proportions. Recording those traits is fundamental to biology and still painfully manual. People now ask chatbots for these facts, but early work found that GPT-4 was only about 49% correct on snake morphological traits (115 descriptions, 24 genera, checked manually), and its answers changed from day to day. For a reference that taxonomists, conservationists, and poison-control centres may rely on, a confident wrong scale count is worse than no answer. Roughly 700 venomous snake species are linked to about 100,000 snakebite deaths a year, so errors in species descriptions have real consequences.

This project does not simply "use AI to extract traits." It builds the machinery that makes AI trait extraction measurable and trustworthy. Over two semesters the team will (1) build a reproducible benchmark that measures what chatbots know about snake traits across many species, models, and prompt phrasings, scored against expert reference data, and (2) build an extraction pipeline with verification and abstention, so that values are reported only when they can be checked against source text. A queryable trait database and dashboard will show what is known, uncertain, and missing for each species.

The project also produces a foundational dictionary of morphological terms and acronyms, with synonyms and homonyms, extracted from snake species descriptions. No complete dictionary of this kind exists. Snakes come first, with lizards, turtles, etc., and possibly other animals as later extensions.

The headline result the team is aiming for: naive prompting was 49% correct; with verification and abstention the system reports fewer values, and the values it reports are N% correct (N to be measured). Benefits: Dr. Uetz and the Reptile Database gain a measured picture of AI reliability and a path to scalable curation, and students gain experience with reproducible LLM evaluation, a problem every organisation deploying LLMs now faces.

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Section A. Problem Statement

Snake species (about 4,200 of them) are told apart mainly by counting and describing scales: rows around the body (dorsal), down the belly (ventral), and under the tail (subcaudal). Those counts are published in the taxonomic literature as numbers, so answers can be checked automatically. Other traits also distinguish species, but they are scattered across original descriptions, books, and papers as messy free text. The same fact appears in many forms ("233 ventrals," "ventrals 233," "V 233," "233 gastrosteges"), and terminology, synonyms, and acronyms vary between authors and over time.

Three problems follow:

1. **Unknown AI reliability.** Preliminary work found a leading model about 49% correct on snake traits, with day-to-day drift. That was one model, one set of prompts, and a few dozen genera. Nobody knows how accuracy varies across models, phrasings, and species.
2. **Unreliable or unverifiable sources.** Chatbot answers often lack sources, or cite sources that do not exist. The Reptile Database, the reference book used for scoring, has neither sources nor sample sizes for its values, so some disagreements between a model and the book may be errors in the book.
3. **No standard vocabulary.** There is no complete dictionary of morphological terms, synonyms, and homonyms (or of the acronyms used in papers), which blocks reliable extraction and comparison across species and models.

The central hypothesis is that accuracy tracks fame rather than biology: models do well on long-known, often-photographed species and badly on recently described, rarely observed ones. Testing this requires a sampling design stratified by year of description and observation count.

Research questions

1. How much do chatbots know, or how much can they pull from the web (measured by full-text descriptions and by the three scale counts)?
2. Is the data reliable and complete? Are sources correctly cited (for example, with working DOIs)?
3. Do models get terminology right: species names, synonyms, morphological terms?
4. Can chatbots reliably extract data from free text into tables that can be compared across species, chatbots, and models?

Section B. Engineering Design Requirements

Requirements derive from the needs of Dr. Uetz and the Reptile Database, the CS faculty advisor's goals for a two-semester capstone, and the requirement that all results be reproducible.

B.1 Project Goals (Client Needs)

1. Measure how accurately, consistently, and honestly chatbots report animal traits starting with snakes, in a way someone else can reproduce.
2. Identify where AI knowledge of animals mainly reptiles like snakes holds, and where it fails (by model, prompt, and species stratum).

3. Build an extraction pipeline that reports a value only when it is verified against source text, and otherwise abstains.
4. Produce a publishable dictionary of morphological terms and acronyms with synonyms and homonyms.
5. Give Dr. Uetz a queryable trait database and a dashboard showing what is known, uncertain, and missing.

B.2 Design Objectives

1. By the end of Semester 1, a single-command, resumable harness runs a sampled species list against multiple vendors, free and paid models, several prompt phrasings, and repeated runs.
2. Every stored response carries a full provenance record or it does not count as evidence (see B.3).
3. Scoring criteria run independently and are reported separately, with no single accuracy number hiding behaviour scores.
4. Correct refusal on genuinely undocumented species is never scored as an error.
5. Substantive disagreements with the reference book go to a review queue for Dr. Uetz instead of being marked wrong.
6. By the end of Semester 2, extraction produces structured trait values against a fixed schema, each with its exact source span (no span, no value).
7. The final report lets an outside reader decide which models to trust, on which species, and with what confidence.

B.3 Design Specifications and Constraints

Constraint	Type	Target / Description
Every run is reproducible from a repository checkout	Reproducibility	Prompts, sampler config, reference spreadsheet, scoring thresholds, and analysis code all live in the repo. A result that cannot be regenerated from a checkout is not a result.
Species list is regenerable from a version number	Reproducibility	Sampler output is versioned; Reptile Database snapshot is versioned and recorded.

Constraint	Type	Target / Description
Every response stored with a full record	Data integrity	species_id, species_name, model_name, model_version_string, prompt_id, prompt_text, temperature, run_index, timestamp_utc, raw_response_text, api_cost_usd, sampler_version, error_or_null
Runs are single-command and resumable	Usability	Interrupted runs continue without repeating completed work.
Cost visible before it is spent; cost caps from the start	Cost	Dry-run cost estimate; free tiers for development, paid models for scored runs.
Model versions pinned and recorded	Reproducibility	model_version_string stored so a silent vendor update becomes a finding, not a hole in the data.
API-accessible models only	Ethical / legal	No scraping of consumer chat interfaces.
Missing values recorded explicitly	Data integrity	Parser returns "no value found" instead of a silent zero.
Sample size	Statistical	About 240 species across 12 strata (about 20 per cell). [Confirm cell definitions; see Section F.1.]
Fixed trait schema	Scope	10-15 traits for v1, agreed with Dr. Uetz and frozen by Week 2; later changes go to a v2 backlog.
Source grounding	Reliability	Every extracted value carries its source span; a value whose span does not contain it is a fabrication.

Constraint	Type	Target / Description
Schedule	Schedule	Complete within tum semesters.

B.4 Codes and Standards

No legal code governs this work directly. The team will follow:

Terms of service and rate limits of each model vendor API, and any Reptile Database access route (API, export, or database dump).

- Reptile Database data-use terms and the licensing of the ~25,000 full-text PDFs in Dr. Uetz's library.
- Citation practice for sources, including DOIs where available.
- An open-source licence for the code and dictionaries,

Section C. Scope of Work

The team will build a reproducible measurement pipeline and a verified trait-extraction pipeline for snake species descriptions, text only. The project follows an agile approach with regular check-ins with the faculty advisor and Dr. Uetz. The benchmark pipeline has five components with defined interfaces:

#	Component	Input	Output
1	Sampler	Reptile Database	Versioned, regenerable species list
2	Harness	Species list, prompt set, model set	Raw responses with full provenance
3	Parser	Raw responses	Structured values, or an explicit "no value found"
4	Scorer	Structured values and reference data	Per-response scores on each criterion
5	Reporter	Scores	Accuracy by model, prompt, and species stratum

The extraction platform builds on these with three further components: **A. Extraction** (schema-constrained values with source spans), **B. Verification and abstention** (domain constraints, source grounding, cross-source consistency), and **C. Trait database and dashboard**.

In scope

- Text only: species descriptions and published literature.
- A bounded starting set (the 24 snake genera already studied, then outward).
- A fixed trait schema developed with Dr. Uetz, not invented by the team.
- Verification, confidence, abstention, and coverage reporting.
- A dashboard, and the term and acronym dictionaries.

Out of scope

- DNA, phylogenetics, images, and computer vision (separate projects; image work could be revisited only if collaboration with Nan Xi is confirmed).
- Training or fine-tuning models.
- Covering all 12,000 species (the pipeline should scale; the deliverable does not have to).
- Beating a benchmark: the contribution is trustworthy output, not the highest score.
- Scraping consumer chat interfaces.

C.1 Deliverables

Technical deliverables

- Sampler producing a versioned species list, stratified by year of description (linear model over every year) and observation count (0-10 / 11-100 / 101-1,000 / 1,000+), plus the reference spreadsheet loaded into a queryable database.
- Harness with API integration, rate limiting, retries, caching, and cost control.
- Parser and Scorer with independent criteria (below) and a disagreement queue.
- Findings report with breakdowns by model, prompt, and species stratum, including how often each model correctly declines.
- Fixed trait schema v1 and the extraction pipeline with source spans.
- Verification checks, confidence scoring, and the abstention rule.
- Trait database and species browser with provenance display and coverage maps.
- Term dictionary and acronym dictionary, mapped to each other, with synonyms and homonyms (Appendix 3).

- Documentation and handoff materials.

Scoring criteria (each an independent check)

- Counts: value in the reference range; overlap if the model gave a range; plausibility; sex-specificity (subcaudal counts differ by sex).
- Behaviour, scored apart from correctness: refusal, hedging language, self-consistency across repeated identical prompts.
- Sources: whether a source was offered, whether it exists, and how old it is.

Academic deliverables

- Team contract and project proposal.
- Semester 1 demo: working pipeline on 24 genera with measured accuracy.
- Final design report, Capstone Expo poster and demonstration.

C.2 Milestones

Milestone	Description	Target
Data access confirmed	Reptile Database access route, PDF library terms, and the 115 checked descriptions secured. Trait schema v1 agreed with Dr. Uetz. Repo, environment, roles.	Sem 1, Weeks 1-2
End-to-end checkpoint	One trait, one genus, end to end, with spans (ugly is fine)	Sem 1, Weeks 3-4
Extraction on checked set	Extraction across the full trait schema on the checked set	Sem 1, Weeks 5-7
Verification checks 1 and 2	Domain constraints and source grounding	Sem 1, Weeks 8-10
Confidence and abstention	Confidence scoring, abstention rule, first accuracy numbers versus the 49%	Sem 1, Weeks 11-13
Semester 1 demo	Working pipeline on 24 genera with measured accuracy	Sem 1, Week 14
Cross-source consistency	Check 3; scale beyond the initial genera	Sem 2, Weeks 1-3

Milestone	Description	Target
Trait database and species browser	Queryable store and browser	Sem 2, Weeks 4-6
Coverage maps and gap reports	Reptile Database cross-linking	Sem 2, Weeks 7-9
Full evaluation	Error taxonomy, calibration analysis	Sem 2, Weeks 10-12
Documentation and Expo	Handoff and Capstone Expo	Sem 2, Weeks 13-14

C.3 Resources

Data

- Reptile Database: taxonomy and records for 12,000+ species, species year of description (maintained by Dr. Uetz).
- All snake species descriptions (All_snake_descriptions_RDB_Sep_2026.zip): tab-delimited, two fields (species name, description), free text with line feeds and references.
- About 25,000 full-text PDFs in Dr. Uetz's library, including original species descriptions.
- The 115 manually checked descriptions across 24 snake genera (Kalashetti & Uetz, in prep.). This is the highest-value item and a ready-made gold standard. Without it, the first month goes into building a gold set by hand.

Software: Python, vendor LLM APIs (free and paid tiers), a PDF-to-text converter, a database for the trait store and reference data, GitHub for version control, and a task tracker.

Human resources: a team of four or five students; Dr. Thomson-McInnes (weekly meetings, framing of verification and abstention); Dr. Uetz (data, schema conversation in Weeks 1-2, domain sanity check *every so often*, review of the disagreement queue in batches).

Budget: see Section I.

C.4 Risks and Mitigation

Risk	Mitigation
Data access is slow	Ask first. Until it lands, work from the 115 descriptions or public Reptile Database pages.
PDF text extraction is messy	Expected and budgeted. Start with clean text, add PDFs later.
Team over-invests in the LLM	Out-of-scope list and the Weeks 3-4 checkpoint.
Trait schema churns	Freeze v1 with Dr. Uetz by Week 2; later changes go to a v2 backlog.
The 49% baseline is not reproducible	Re-measure it; that becomes a result.
Vendors update or deprecate models without notice	Pin and record model_version_string; treat mid-study changes as findings.
API cost overruns	Cost caps and dry-run estimates built into the harness from day one.
Reference book errors	Disagreement queue reviewed by Dr. Uetz rather than automatic penalties.

Section D. Concept Generation

Three concepts for the overall approach are considered.

Concept 1: Benchmark-Only Study of Out-of-the-Box Chatbots

Sample about 240 species, ask multiple chatbots (several phrasings, repeated runs) for the three scale counts and for descriptions, and score the answers against the Reptile Database. No attempt is made to fix the models' errors.

Pros

- Directly answers research questions 1-3 and re-measures the 49% baseline.
- Scored automatically because scale counts are numbers.
- Most achievable in one semester.

Cons

- Measures the problem without improving trust in the output.
- Reference book has no sources or sample sizes, so some disagreements are the book's fault.
- Findings tied to specific model versions that may change.

Sub-problems: sampling design, harness cost control, parsing messy answers, explicit "no value" handling, disagreement queue.

Concept 2: Schema-Constrained Extraction with Verification and Abstention

Give real species descriptions to a model and extract values against a fixed trait schema, each with its exact source span. Check every value three ways (domain constraints, source grounding, cross-source consistency). Withhold failed or low-confidence values as "unknown."

Pros

- Targets the real need: trustworthy values for a reference database.
- Every value is traceable to text; fabrication is detectable.
- Coverage grows only as fast as values can be verified.

Cons

- Needs the fixed schema and gold set early.
- PDF-to-text extraction is messy.
- Greater risk of over-investing in LLM quality rather than reliability.

Sub-problems: schema design with Dr. Uetz, span capture, term and unit normalization ("233 ventrals / V 233 / 233 gastrosteges"), confidence calibration.

Concept 3: RAG / GraphRAG over Reptile Database Descriptions

Index the descriptions and literature and let a retrieval system feed relevant text to a model, with or without a knowledge graph of species, traits, and terms.

Pros

- Could reduce hallucination by grounding answers in retrieved text.
- Natural fit with a trait and term dictionary.

Cons

- Largest engineering effort and the hardest to reproduce exactly.
- Retrieval quality becomes another variable to evaluate.
- Depends on the corpus and licensing being settled.

Sub-problems: corpus preparation, chunking, retrieval evaluation, graph construction.

Section E. Concept Evaluation and Selection

E.1 Selection Criteria and Metrics

Criterion	Metric	Weight	Rationale
Trustworthiness of output	Does the approach measure or improve reliability of reported values? (1 = low, 3 = high)	3	The core need: a confident wrong scale count is worse than no answer
Reproducibility	Can results be regenerated exactly when a component is swapped? (1 = hard, 3 = easy)	3	A result that cannot be regenerated is not a result
Feasibility in two semesters	Development effort (1 = high, 3 = low)	2	Student time limits
Scalability to other taxa	Extends to lizards, turtles, insects (1 = poor, 3 = excellent)	1	Future expansion without major refactoring
Risk of failure	Likelihood of blockers (1 = high risk, 3 = low risk)	2	Data access, PDF quality, cost

E.2 Decision Matrix

Scores below are only temporary and will later be confirmed by the team and faculty.

Concept	Trustworthiness (3)	Reproducibility (3)	Feasibility (2)	Scalability (2)	Risk (2)	Weighted Score
1. Benchmark only	2	3	3	2	3	*23 + 33 + 32 + 22* + 3*2 = 31
2. Extraction + verification	3	3	2	3	2	*33 + 33 + 22 + 32* + 2*2 = 32
3. RAG / GraphRAG	2	2	1	3	1	*23 + 23 + 12 + 32* + 1*2 = 22

Concepts 1 and 2 score closely, and they are complementary rather than competing: the benchmark supplies the baseline and the gold-standard comparisons that the verification layer must beat.

E.3 Selected Design

The proposed design is a **phased combination of Concepts 1 and 2**:

- **Semester 1:** build the reproducible benchmark pipeline (Concept 1) and use it, plus the 115 checked descriptions, to establish the baseline and the first extraction results.
- **Semester 2:** add verification, abstention, the trait database, and the dashboard (Concept 2).
- **Deferred:** RAG/GraphRAG (Concept 3) is a possible next step once the team sees what the internet and chatbots already provide. It is not committed.

Rationale: this keeps a single unified plan (the earlier draft had two overlapping breakdowns), delivers a publishable finding by the end of Semester 1, and keeps verification as the heart of the project.

Section F. Design Methodology

F.1 Computational Methods

- **Sampling.** Pull the species list and year of description from the Reptile Database, and research-grade observation counts from the API. Stratify on year of description (modelled as a linear trend over every year; sampling bins [TBD]) and observation count (0-10 / 11-100 / 101-1,000 / 1,000+). The target is about 20 species per cell, roughly 240 species. [The earlier draft says "twelve cells"; four observation bins times three year bins gives twelve, so three year bins must be defined with Dr. Uetz.]

- **Hypothesis.** Accuracy tracks fame rather than biology.
- **Harness.** Multiple vendors, free and paid models, several phrasings per question, repeated identical runs to measure drift. Two question types: the three scale counts and open-ended descriptions.
- **Parsing.** Extract structured values from free text, recording "no value found" explicitly.
- **Statistics.** Compare models, prompts, and strata with tests suited to the data [methods TBD with advisor], reporting uncertainty rather than a single accuracy number.
- **Dictionary building.** See Appendix 3.

F.2 Experimental Methods

- Repeat identical prompts across days to measure drift and self-consistency.
- Hold out species with no usable description to test correct abstention versus confident-wrong rates.
- Re-measure the 49% baseline on the checked set.
- Compare answers when a source is requested versus not requested.

F.3 Architecture and High-Level Design

Sampler, Harness, Parser, Scorer, and Reporter run as separable stages with defined interfaces, so any one (the model, a prompt, a dataset) can be switched out and the run repeated exactly. The extraction stack adds the schema-constrained extractor, the three verification checks, and the trait store with a dashboard. Everything lives in a single repository: prompts, sampler config, reference spreadsheet, scoring thresholds, and analysis code.

F.4 Verification and Validation Procedure

Question	Measured how
Are extracted values correct?	Against the 115 manually checked descriptions; report accuracy per trait type
Is the provenance real?	Does the cited span actually contain the value? Measurable directly

Question	Measured how
Does it abstain when it should?	Hold out species with no usable description; correct-abstention rate versus confident-wrong rate
Is confidence meaningful?	Do high-confidence values have higher accuracy than low-confidence ones? Plot it
Does it beat the naive baseline?	The 49% is the baseline; state the number both ways (fewer values, higher accuracy)
Are sources real?	Check whether cited sources exist (for example, DOI resolution) and their age

F.5 Validation Procedure

Validation happens in Semester 2: a demonstration to Dr. Uetz and Dr. Thomson-McInnes, a review of the disagreement queue and a sample of extracted values by Dr. Uetz, and a readability check of the findings report by someone outside the team. Questions for Dr. Uetz are batched rather than sent one at a time.

Section G. Results and Design Details

This proposal-stage section describes the results the project will report and the final design. No results exist yet; table entries are placeholders to be completed as data arrives.

G.1 Modeling Results

Planned: data flow diagram (Sampler to Reporter), system architecture diagram, and the trait schema v1 with value types and units.

G.2 Experimental Results

Test	Metric	Goal	Result
Baseline re-measurement	Accuracy on 115 checked descriptions	Reproduce or refute 49%	Pending

Test	Metric	Goal	Result
Scale-count accuracy by model	Value in range / range overlap	Report per model	Pending
Accuracy by stratum	Year of description, observation count	Test "fame, not biology"	Pending
Drift	Self-consistency across repeated runs	Measure per model	Pending
Correct abstention	Refusal on undocumented species	Report rate	Pending
Source reality	Offered, exists, age	Report per model	Pending
Verified extraction	Accuracy of reported values vs coverage	Beat 49% on reported values	Pending

G.3 Prototyping and Testing Results

Planned: Weeks 3-4 end-to-end prototype (one trait, one genus, with spans), then expansion to the full schema on the checked set.

G.4 Final Design Details and Specifications

Specification	Implementation
Species list	Versioned, regenerable from a version number
Reference data	Spreadsheet transcribed from the reference book and loaded into a queryable database
Provenance record	All 13 fields listed in B.3 stored per response
Runs	Single command, resumable, cached, with cost visible before spending
Scoring	Independent criteria reported separately; disagreement queue for Dr. Uetz
Extraction	Fixed schema, exact source span per value, curated trait list only

Specification	Implementation
Abstention	Withhold values that fail checks or fall below the confidence threshold
Trait store	Species, traits, values, confidence, provenance; cross-linked to Reptile Database entries
Dashboard	Species browser, known/uncertain/missing view, coverage maps
Documentation	README, setup and re-run instructions, trait schema, scoring definitions

Section H. Societal Impacts of Design

H.1 Public Health, Safety, and Welfare

About 700 venomous snake species are linked to roughly 100,000 snakebite deaths a year. Wrong species descriptions can mislead identification and treatment decisions, so abstention and explicit uncertainty are safety features. The system is a research and curation aid, not a clinical tool, and the report will say so.

H.2 Societal Impacts

A measured account of where AI is and is not reliable helps scientists, conservationists, and the public decide how much to trust chatbot answers about wildlife. An open morphological dictionary lowers barriers for researchers and students.

H.3 Political and Regulatory Impacts

No personal data is involved. The main obligations are respecting vendor and data-source terms, licensing of the PDF library, and attribution of sources. Publishing dictionaries and code requires licensing decisions (TBD).

H.4 Economic Impacts

The pipeline could reduce the manual labour of curating species traits and may apply across taxonomic groups. Cost caps keep evaluation affordable for a small team.

H.5 Environmental Impacts

LLM runs consume compute and energy. Caching, never repeating an identical paid prompt, and free-tier development limit unnecessary usage. Better species information may support conservation work.

H.6 Global Impacts

Reptile diversity and snakebite burden are global. Open dictionaries and open methods can be reused internationally, and the approach is designed to extend to other taxa.

H.7 Ethical Considerations

- Honest reporting of failures, including when the reference book may be the source of disagreement.
- Correct refusal is never penalized as an error.
- Proper attribution of sources, including the original describers and the Reptile Database.
- Stating limitations, including that consumer chat interfaces were not tested.
- Not presenting AI output as expert-verified when it is not.

Section I. Cost Analysis

Most costs are API usage and hosting. Figures below are placeholders to be replaced with real estimates once prompt counts and model prices are fixed.

I.1 Current Costs

Resource	Provider	Unit Cost	Quantity	Total	Notes
Free-tier LLM APIs (development)	Various vendors	$0.00	-	$0.00	Development and testing
Paid LLM APIs (scored runs)	Various vendors	[TBD]	[TBD]	[TBD]	Estimated as species x prompts x runs x models x tokens; capped in the harness
Repository and task tracking	GitHub or equivalent	$0.00	-	$0.00	Free tier
Database and hosting for the dashboard	[TBD]	[TBD]	1	[TBD]	University or free-tier resources

Resource	Provider	Unit Cost	Quantity	Total	Notes
Expo materials	N/A	[TBD]	1	[TBD]	Poster and printing

I.2 Anticipated Future Costs

Category	Estimated Cost	Description
Additional paid model runs	[TBD]	Repeats after a vendor changes model versions
Extension to lizards or insects	[TBD]	Only if pursued

I.3 Cost Summary

The project's largest variable cost is paid API usage for the scored runs. The harness will estimate cost before a run starts, enforce hard caps, and avoid paying twice for an identical prompt to an identical model version. [Total budget TBD with Dr. Thomson-McInnes.]

Section J. Conclusions and Recommendations

To be completed at the end of the project.

Lessons learned: [TBD]

Obstacles overcome: [TBD]

Summary of final design features: [TBD]

Future recommendations (candidates, none promised):

1. Give real species descriptions to a model to extract, normalize, and define specialized vocabulary, producing a machine-readable dictionary of snake morphology.
2. Test whether a model can find a labelled drawing that illustrates each term (would require resolving the out-of-scope status of images).
3. Test whether the approach carries over to lizards, turtles, and insects.

Appendix 1: Project Timeline

Semester 1 and Semester 2 each run 14 weeks (see milestones in C.2). The chart is still to be produced once the term calendar and course deliverable dates are known.

Period	Work
Sem 1, Weeks 1-2	Data access, trait schema v1, repo, environment, roles
Sem 1, Weeks 3-4	End-to-end checkpoint
Sem 1, Weeks 5-7	Full-schema extraction on checked set
Sem 1, Weeks 8-10	Verification checks 1 and 2
Sem 1, Weeks 11-13	Confidence scoring, abstention rule, first accuracy numbers
Sem 1, Week 14	Semester 1 demo and report
Sem 2, Weeks 1-3	Cross-source consistency, scale beyond initial genera
Sem 2, Weeks 4-6	Trait database and species browser
Sem 2, Weeks 7-9	Coverage maps, gap reports, Reptile Database cross-linking
Sem 2, Weeks 10-12	Full evaluation, error taxonomy, calibration analysis
Sem 2, Weeks 13-14	Documentation, handoff, Capstone Expo

Appendix 2: Team Contract (Team Organization)

To be completed once the team is assigned.

Step 1: Team members, strengths, and contact information [TBD]

Step 2: Culture goals, actions, and warning signs [TBD]

Step 3: Meetings

Meeting	Participants	Frequency	Goal
Team working sessions	Students	Whenever possible	Day-to-day progress
Advisor meeting	Students and Dr. Thomson-McInnes	Weekly	Framing of verification and abstention; help when extraction quality tempts the team more than reliability
Domain check-in	Students and Dr. Uetz	Not weekly: schema conversation in Weeks 1-2, then a domain sanity check once or twice a semester, plus batched questions	Schema, data, disagreement queue review

Step 4: Roles. Everyone writes and reviews code; each area has one person answerable for it.

Role	Area	Responsibilities
Sampling and data lead	Sampler, reference data	Species list from Reptile Database and iNaturalist, stratification, reference spreadsheet and database, schema design with Dr. Uetz, PDF-to-text handling
Harness lead	Harness	API integration, rate limits, retries, caching, cost control, provenance records
Parsing and scoring lead	Parser, Scorer, Verification	Structured value parsing, independent scoring criteria, disagreement queue, verification checks, abstention rule
Analysis and interface lead	Reporter, Dashboard	Statistics, findings report, trait database, species browser, coverage maps

Role	Area	Responsibilities
Fifth member (if assigned)	Evaluation	Gold-set management, metrics, error taxonomy

Step 5: Agreement/ Jeremiah Summers, Daniel Zerrusen, Andrew Dickerson, Amro Rawi

Appendix 3: Term and Acronym Dictionary Workflow

1. **Examine the corpus.** All_snake_descriptions_RDB_Sep_2026.zip contains descriptions for almost all snake species: tab-delimited with two fields (species name, description). The text is messy: line feeds, multiple paragraphs, very short entries, some with only a reference. A paragraph usually ends with its source reference (for example, [Maddock et al. 2015]). The type species of a genus often also carries a genus description.
2. **Build a term dictionary.** Extract all nouns, including compound nouns (for example, "upper head-shields," "upper labials"), in singular and plural form (subocular, suboculars). Use Acanthophis antarcticus as the worked example.
3. **Organize terms.** Group into primary terms, synonyms, and homonyms using LLMs with carefully designed prompts, following the approach in the pilot paper by Yaa Darko, a previous VCU student.
4. **Repeat for acronyms.** The Darko pilot used 50 papers; this project expands to thousands. Requires a PDF-to-text converter and a set of original-description PDFs provided by Dr. Uetz (example: ZooKeys article 186365, Materials and Methods, Morphological data evaluation).
5. **Map terms to acronyms.** Not every term has an acronym. Different papers use different mappings, acronyms, and terms, which creates the synonym and homonym problem.

Output: two publishable dictionaries (terms and acronyms) with mappings, foundational for the field. The method can apply to any taxonomic group: snakes first, then lizards, turtles, and possibly others.

Appendix 4: Open Questions for Faculty and Domain Partner

Ask Dr. Uetz for these three items first, in this order:

1. Access route and terms for the Reptile Database (API, export, or database dump).

2. What the team may use from the ~25,000 full-text PDFs, and how they are delivered.
3. The 115 manually checked descriptions (24 genera) in usable form.

Other open decisions:

- Year-of-description bins for the 12 sampling cells.
- Whether to involve Nan Xi (and images), given his own capstone commitments. Peter Uetz to email.
- Team size (four or five) and final role assignments.
- Whether RAG or GraphRAG is pursued or deferred.
- Budget cap and licence for code and dictionaries.
- Course deliverable dates for the chart.

References

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[4] iNaturalist API v1 documentation. https://api.inaturalist.org/v1/docs/

[5] ZooKeys article 186365 (example of morphological data evaluation). https://zookeys.pensoft.net/article/186365/

[6] Maddock et al. (2015) and Boulenger (1896): example description sources cited in the draft.